



PRAKTIKUM

Pemrograman Dasar

Praktikum Kelas A

Asisten Praktikum

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**Program Studi Pendidikan Teknologi
Informasi
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BAB 1

INPUT, PROSES, OUTPUT

1.6 Data dan Analisis Hasil Percobaan

1.6.1 Output

1. Jalankan Kelas Output1-Output4 dan benahi jika menemukan kesalahan!



The image consists of two screenshots of an IDE, likely IntelliJ IDEA, showing Java code and terminal output. The background of the IDE is a dark theme with a manga-style illustration.

Top Screenshot:

- File Explorer: `inputjava 2` and `Outputjava 1` are open.
- Code Editor: `Output.java` is open. The code is:

```
public class Output {  
    public static void main(String[] args) {  
        System.out.println(x: "Hello World");  
    }  
}  
//Hashbi Abdillah  
//265150600111022
```
- Terminal: Shows the command `java Output.java` and the output `Hello World`.

Bottom Screenshot:

- File Explorer: `prak2.java 2`, `variable.java 2`, `input.java 2`, and `Output2.java` are open.
- Code Editor: `Output2.java` is open. The code is:

```
public class Output2 {  
    public static void main(String[] args) {  
        System.out.println(x: x: "Belajar");  
        System.out.println("pemrograman " + "Java");  
    }  
}  
//Hashbi Abdillah  
//265150600111022
```
- Terminal: Shows the command `java Output2.java` and the output `Belajar` and `pemrograman Java`.

```
Output3.java > Output3 > main(String[])
1 public class Output3 {
2     public static void main(String[] args) {
3         System.out.printf(format: format: "%s%c\n", ...args: ...args: "Pemrograman
4         java", '|');
5         System.out.printf(format: format: "%30s%c\n", ...args: ...args: "Pemrograman
6         java", '|');
7         System.out.printf(format: format: "%-30s%c\n", ...args: ...args: "Pemrograman
8         java", '|');
9         System.out.printf(format: format: "%30.5s%c\n", ...args: ...args: "Pemrograman
10        java", '|');
11    }
12 }
13 //Hashbi Abdillah
14 //265150600111022

Problems 9 Output Debug Console Terminal Ports
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdas-sem1\g\mod1> java Output3.java
Pemrograman java|
Pemrograman java|
Pemrograman java|
Pemrograman java|
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdas-sem1\g\mod1>

Output4.java > Output4 > main(String[])
1 public class Output4 {
2     public static void main(String[] args) {
3         float x = 7654.123456789f;
4         System.out.printf(format: format: "%d %d %8d\n", ...args: ...args: 1234, -567, 8910)
5         System.out.printf(format: format: "%d %d %8d\n", ...args: ...args: 1234, -567, 8910)
6         System.out.printf(format: format: "%f %15f %15.3f\n", x, x, x);
7     }
8 }
9 //Hashbi Abdillah
10 //265150600111022

Problems 9 Output Debug Console Terminal Ports
7654,12 7,654124e+03 7654,124
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdas-sem1\g\mod1> java Output4.java
1234 -567 8910
1234 -567 +8910
7654,123535 7654,123535 7654,124
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdas-sem1\g\mod1>
```

2. error, karena terdapat 2 string tapi tidak digabungkan
- 3.a. line 3 mencetak line biasa secara normal, %s mencetak string dan %c mencetak karakter.
- 3.b. line 4 mencetak line rata kanan dengan alokasi 30 karakter karena menggunakan %30s dan %c mencetak karakter untuk penutup
- 3.c. line 5 adalah line 4, dibalik menjadi line rata kiri karena menggunakan %-30s
- 3.d. memotong hanya 5 karakter pertama karena ditambahkan .5, dicetak rata kanan

4. error karena jika menggunakan kutip 1 bukan bertipe string melainkan char yang dimana char hanya bernilai tidak boleh lebih dari 1 karakter

5. mencetak 8 karakter awal pada string dengan 30 space karakter, dengan format rata kiri dengan '|' sebagai penutup di sebelah kanan

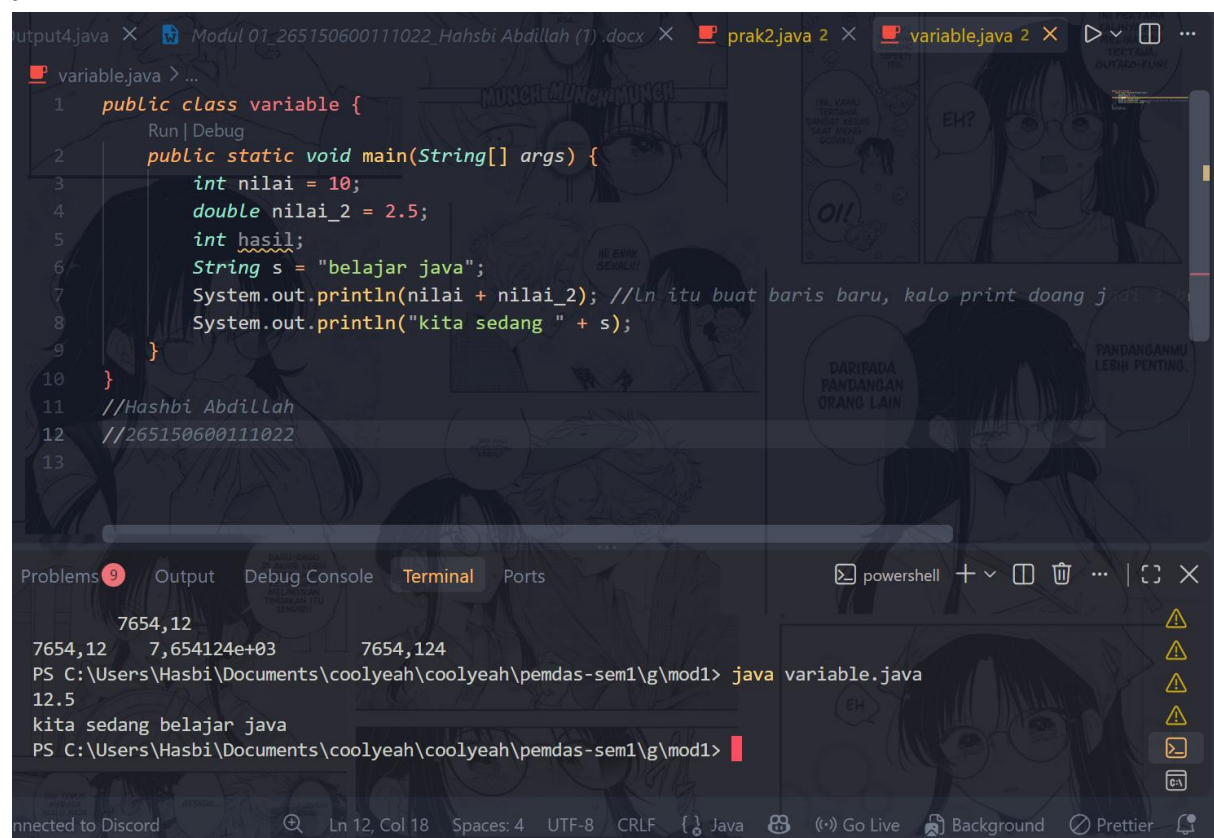
6. `System.out.printf("%15.2f\n", x);`

7. Kode tersebut mencetak angka dengan tiga format berbeda: %g (ringkas), %15e (eksponensial), dan %15.3f (desimal).

8. Kode ini menunjukkan cara mengatur tampilan angka saat dicetak, %d: menampilkan angka bulat, %f: menampilkan angka pecahan (desimal), %g: menampilkan angka secara ringkas, %e: menampilkan angka dalam bentuk eksponen (pangkat), %3d / %15f: mengatur lebar spasi angka, %.3f: membatasi berapa angka di belakang koma.

1.6.2 Variabel

9.



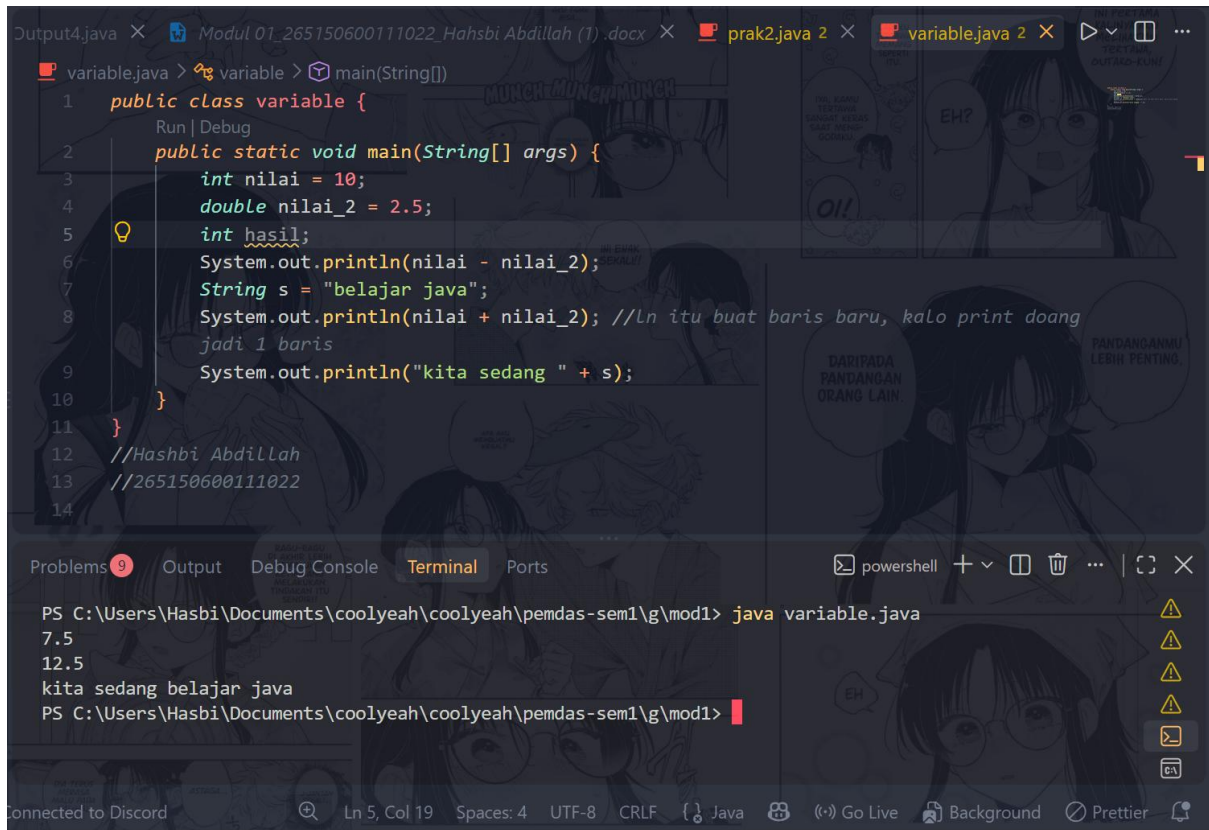
The screenshot shows an IDE with a Java file named `variable.java` and its output in the terminal. The code defines a class `variable` with a `main` method. It declares variables `nilai` (int, 10), `nilai_2` (double, 2.5), `hasil` (int), and `s` (String, "belajar java"). It then prints the sum of `nilai` and `nilai_2` followed by `s` on the same line, and then `s` on a new line. The terminal output shows the results of these operations.

```
1 public class variable {
2     public static void main(String[] args) {
3         int nilai = 10;
4         double nilai_2 = 2.5;
5         int hasil;
6         String s = "belajar java";
7         System.out.println(nilai + nilai_2); //ln itu buat baris baru, kalo print doang j...
8         System.out.println("kita sedang " + s);
9     }
10 }
11 //Hashbi Abdillah
12 //265150600111022
13
```

Terminal Output:

```
7654,12
7654,12 7,654124e+03 7654,124
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1> java variable.java
12.5
kita sedang belajar java
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1>
```

10.



The screenshot shows an IDE with a Java file named `variable.java`. The code defines a `main` method that calculates the difference between two numbers, prints it, and then concatenates it with a string. The terminal output shows the results of these operations.

```
variable.java > variable > main(String[])
1 public class variable {
2     Run | Debug
3     public static void main(String[] args) {
4         int nilai = 10;
5         double nilai_2 = 2.5;
6         int hasil;
7         System.out.println(nilai - nilai_2);
8         String s = "belajar java";
9         System.out.println(nilai + nilai_2); //Ln itu buat baris baru, kalo print doang
10        jadi 1 baris
11        System.out.println("kita sedang " + s);
12    }
13    //Hashbi Abdillah
14    //265150600111022
```

Problems 9 Output Debug Console Terminal Ports

```
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdass-sem1\g\mod1> java variable.java
7.5
12.5
kita sedang belajar java
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdass-sem1\g\mod1>
```

Connected to Discord Ln 5, Col 19 Spaces: 4 UTF-8 CRLF Java Go Live Background Prettier

11. error karena char hanya bisa menyimpan 1 nilai karakter

The image shows a screenshot of a code editor with a dark theme. The editor has several tabs at the top: 'Output4.java', 'Modul 01_265150600111022_Hahsbi Abdillah (1).docx', 'prak2.java 2', 'variable.java', and a Java icon. The 'variable.java' tab is active, showing the following code:

```
1 public class variable {
2     public static void main(String[] args) {
3
4         double nilai_2 = 2.5;
5         int nilai_3 = 100;
6         int hasil = nilai_3 / 2;
7         System.out.println(nilai - nilai_2);
8         String s = "belajar java";
9         System.out.println(nilai + nilai_2); //ln itu buat baris baru, kalo print doang
10                                             jadi 1 baris
11         System.out.println("kita sedang " + s);
12         System.out.println(hasil);
13         // char ch = 'belajar java';
14     }
15 //Hashbi Abdillah
16 //265150600111022
```

Below the code editor is a panel with tabs for 'Problems' (7), 'Output', 'Debug Console', 'Terminal', and 'Ports'. The 'Terminal' tab is active, showing the command prompt output:

```
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1> java variable.java
7.5
12.5
kita sedang belajar java
50
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1>
```

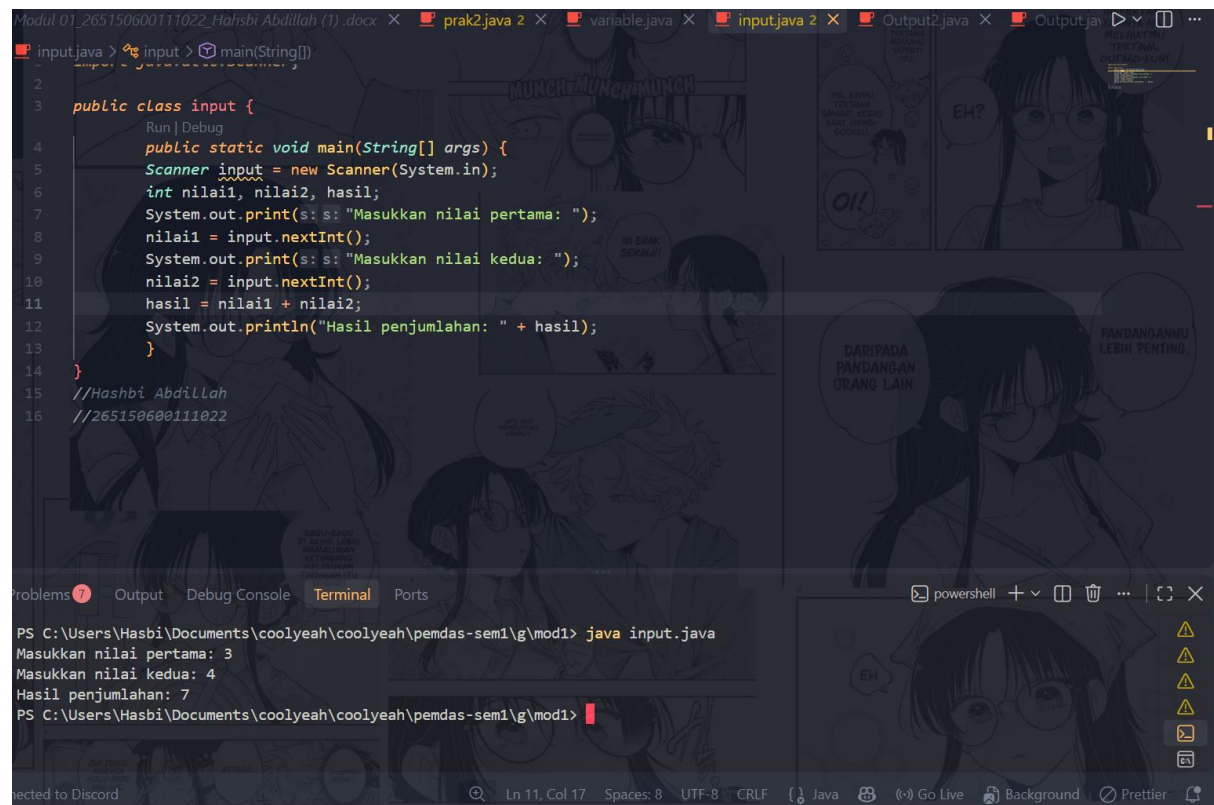
The background of the entire image is a faint, repeating pattern of anime-style illustrations of a girl with glasses and pigtails, with various speech bubbles and text overlays.

The image is a screenshot of a code editor with a dark theme. The main editor window displays a Java file named 'variable.java'. The code defines a public class 'variable' with a static method 'main' that takes an array of strings as an argument. Inside the 'main' method, it declares and initializes 'nilai_3' to 100, calculates 'hasil' as 'nilai_3' divided by 2, and prints 'nilai - nilai_2'. It then sets a string 's' to 'belajar java' and prints 'nilai + nilai_2' with a comment indicating it should be on a new line. Next, it prints 'kita sedang ' followed by 's', then 'nilai_3 / 2' with a lightbulb icon above it, and finally prints the character 'c' from the string 'belajar java'. The file path at the bottom of the editor is '//Hashbi Abdillah //265150600111022'. To the right of the code editor is a vertical panel with tabs for 'problems', 'output', 'debug console', 'terminal', and 'ports'. The 'terminal' tab is active, showing the command prompt output: '7.5', '12.5', 'kita sedang belajar java', '50', '50', and the current directory path 'PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1>'. The background of the IDE is decorated with a faint anime-style illustration of a girl with glasses and pigtails.

14. no 12 proses perhitungan dilakukan dengan memasukkan nilai ke variabel hasil, sedangkan no 13 perhitungan dilakukan sebelum mencetak hasil print tanpa menggunakan variabel hasil

1.6.3 Input

15.



```
Modul 01_265150600111022_Hahsbi Abdillah (1) .docx X prak2.java 2 X variable.java X input.java 2 X Output2.java X Outputja X ...
input.java > Run | Debug
main(String[])
2
3 public class input {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6         int nilai1, nilai2, hasil;
7         System.out.print(s: s: "Masukkan nilai pertama: ");
8         nilai1 = input.nextInt();
9         System.out.print(s: s: "Masukkan nilai kedua: ");
10        nilai2 = input.nextInt();
11        hasil = nilai1 + nilai2;
12        System.out.println("Hasil penjumlahan: " + hasil);
13    }
14 }
15 //Hashbi Abdillah
16 //265150600111022

Problems 7 Output Debug Console Terminal Ports
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1> java input.java
Masukkan nilai pertama: 3
Masukkan nilai kedua: 4
Hasil penjumlahan: 7
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemas-sem1\g\mod1>

Ln 11, Col 17 Spaces: 8 UTF-8 CRLF Java Go Live Background Prettier
```

16. Error karena tipe data yang di declare berupa integer

17.



The screenshot shows an IDE with a Java file named `input.java`. The code is as follows:

```
1 import java.util.Scanner;
2
3 public class input {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6         int nilai1, nilai2, hasil;
7         String nama, nim;
8         System.out.print(s: s: "Masukkan nilai pertama: ");
9         nilai1 = input.nextInt();
10        System.out.print(s: s: "Masukkan nilai kedua: ");
11        nilai2 = input.nextInt();
12        input.nextLine();
13        System.out.print(s: s: "Masukkan nama: ");
14        nama = input.nextLine();
15        System.out.print(s: s: "Masukkan nim: ");
16        nim = input.nextLine();
17        hasil = nilai1 + nilai2;
18        System.out.println("Hasil penjumlahan: " + hasil);
19        System.out.println("nama: " + nama);
20        System.out.println("nim: " + nim);
21    }
22 }
```

The IDE's output window shows the execution results:

```
Hasil penjumlahan: 11
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdass-1\mod1> java input.java
Masukkan nilai pertama: 5
Masukkan nilai kedua: 4
Masukkan nama: hashbi abdillah
Masukkan nim: 265150600111022
Hasil penjumlahan: 9
nama: hashbi abdillah
nim: 265150600111022
PS C:\Users\Hasbi\Documents\coolyeah\coolyeah\pemdass-1\mod1>
```

1.7 Tugas Praktikum

1.7 Tugas Praktikum

1. Buatlah program dengan tampilan sebagai berikut.
Masukkan operator pertama : 3
Masukkan operator kedua : 2
Hasil penjumlahan : 5
Hasil pengurangan : 1
Hasil perkalian : 6
Hasil pembagian : 1.5
2. Buatlah program untuk menghitung pemakaian daya listrik dirumah tangga secara sederhana. Tampilan program sebagai berikut.

Program penghitung pemakaian listrik sederhana

Masukkan Nama : Bpk Asisten

Kelurahan : Java

Masukkan posisi awal Kwh Meter : 8000

Masukkan posisi akhir Khw Meter : 9000

Masukkan biaya beban saat ini : 140

Masukkan PPJ (dalam persen) : 10

=====PLN Java=====

Nama : Bpk Asisten

Kelurahan : Java

Pemakaian bulan ini : 1000 Kwh Meter

Tarif Listrik : Rp 140000,-

PPJ 10% : Rp 14000,-

Total Bayar : Rp 154000,-

=====

Praktikum 1



```
parkir.java 2 x .gitignore U x praktik.java 1 x
g > mod1 > praktik.java > ...
1 import java.util.Scanner;
2
3 public class praktik {
4     public static void main(String[] args) {
5         Scanner input = new Scanner(System.in);
6         int nilai1, nilai2;
7         double hasilJml, hasilKrg, hasilKli, hasilBgi;
8         System.out.print(s: "Masukkan nilai pertama: ");
9         nilai1 = input.nextInt();
10        System.out.print(s: "Masukkan nilai kedua: ");
11        nilai2 = input.nextInt();
12        hasilJml = (double) nilai1 + nilai2;
13        System.out.println("Hasil penjumlahan: " + hasilJml);
14        hasilKrg = (double) nilai1 - nilai2;
15        System.out.println("Hasil penjumlahan: " + hasilKrg);
16        hasilKli = (double) nilai1 * nilai2;
17        System.out.println("Hasil penjumlahan: " + hasilKli);
18        hasilBgi = (double) nilai1 / nilai2;
19        System.out.println("Hasil penjumlahan: " + hasilBgi);
20    }
21 }
22 //Hashbi Abdillah
23 //265150600111022
24

Problems Output Debug Console Terminal Ports
Masukkan nilai pertama: 5
Masukkan nilai kedua: 2
Hasil penjumlahan: 7.0
Hasil penjumlahan: 3.0
Hasil penjumlahan: 10.0
Hasil penjumlahan: 2.5
PS C:\Users\Hasbi\Documents\coolyeah(not main)\coolyeah\pemdass-semester1\mod1>
```

Praktikum 2

```
parkir.java 2 X .gitignore u X prak2.java 1 M X
g > mod1 > prak2.java > prak2 > main(String[])
3 public class prak2 {
4     public static void main(String[] args) {
5         int kwhAwal, kwhAkhir, biayaBeban, ppj, tarifppj, tarif, totPemakaian, totTarif, panjangSisi=20;
6         String nama, kelurahan, garisSisi= "=".repeat(panjangSisi) ;
7         System.out.print(s: "Masukkan Nama: ");
8         nama = input.nextLine();
9         System.out.print(s: "Masukkan kelurahan: ");
10        kelurahan = input.nextLine();
11        System.out.print(s: "Masukkan Kwh Awal: ");
12        kwhAwal = input.nextInt();
13        System.out.print(s: "Masukkan Kwh Akhir: ");
14        kwhAkhir = input.nextInt();
15        System.out.print(s: "Masukkan biaya beban: ");
16        biayaBeban = input.nextInt();
17        System.out.print(s: "Masukkan ppj: ");
18        ppj = input.nextInt();
19        //Hashbi Abdillah
20        //265150600111022
21        totPemakaian = kwhAkhir - kwhAwal;
22        tarif = totPemakaian * biayaBeban;
23        tarifppj= tarif*ppj/100;
24        totTarif= tarif + tarifppj;
25
26
27        String textHeader ="PLN" + kelurahan, header = garisSisi + textHeader + garisSisi, footer="=".repeat(header.length());
28
29        System.out.printf(format: "%s\n", header) ;
30        System.out.println("Nama: " + nama);
31        System.out.println("Kelurahan: " + kelurahan);
32        System.out.println("Total pemakaian bulan ini: " + totPemakaian);
33        System.out.println("Tarif listrik: " + tarif);
34        System.out.printf("PPJ %d%%: " + tarifppj + "%n", ppj);
35        System.out.println("Total bayar: " + totTarif);
36
37        System.out.printf(format: "%s\n", footer);
38    }
39 }
```

Problems 6 Output Debug Console Terminal Ports

```
Masukkan ppj: 10
=====PLNBakwanjagung=====
Nama: udin knalpot bobok
Kelurahan: bakwanjagung
Total pemakaian bulan ini: 1000
Tarif listrik: 140000
PPJ 10%: 14000
Total bayar: 154000
=====
PS C:\Users\Hasbi\Documents\coolyeah(not main)\coolyeah\pemdass-sem1\g\mod1>
```

Java: Ready Connected to Discord HashbiAbdillah (3 days ago) Ln 37, Co